10.3 Enzyme Catalysis

Enzymes are complex protein molecules with three dimensional structures. They catalyse the chemical reaction in living organism. They are often present in colloidal state and extremely specific in catalytic action. Each enzyme produced in a particular living cell can catalyse a particular reaction in the cell.

Some common examples for enzyme catalysis

1) The peptide glycyl L-glutamyl L-tyrosin is hydrolysed by an enzyme called pepsin.
2) The enzyme diastase hydrolyses starch into maltose

$$2(C_6H_{10}O_5)_n + nH_2O \rightarrow nC_{12}H_{22}O_{11}$$

3) The yeast contains the enzyme zymase which converts glucose into ethanol.

$$C_6H_{12}O_6 \rightarrow 2C_2H_5OH + 2CO_2$$

4) The enzyme micoderma aceti oxidises alcohol into acetic acid.

$$C_2H_5OH + O_2 \rightarrow CH_3COOH + H_2O$$

5) The enzyme urease present in soya beans hydrolyses the urea.

$$NH_2\text{-}CO\text{-}NH_2 + H_2O \rightarrow 2NH_3 + CO_2$$

10.3.1 Mechanism of enzyme catalysed reaction

The following mechanism is proposed for the enzyme catalysis

$$E + S \rightleftharpoons ES \rightarrow P + E$$

Where E is the enzyme, S the substrate (reactant), ES represents activated complex and P the products.

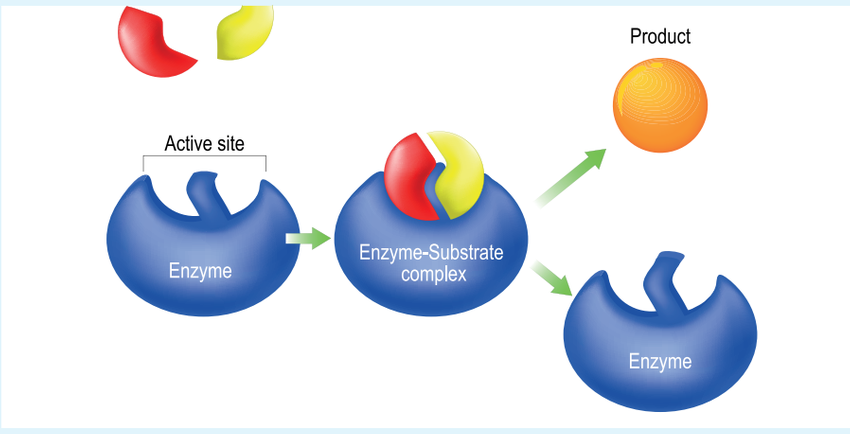


Figure 10.5 Mechanism of Enzyme Catalysis

Enzyme catalysed reaction show certain general special characteristics.

(i) Effective and efficient conversion is the special characteristic of enzyme catalysed reactions. An enzyme may transform a million molecules of reactant into product in a minute.

For eg. $2H_2O_2 \rightarrow 2H_2O + O_2$

For this reaction, the activation energy is 18k cal/mole without a catalyst.

With colloidal platinum as a catalyst the activation energy is 11.7kcal /mole

XII U10-Surface Chemistry.indd 83 16-12-2022 19:40:22